

CSE-207

Data Structures

Section 5

Date: 23/06/22

- 1 Describe different types of data structures with example
- 2 For the following set of data use binary search to find the value of a number called
ITEM=110. A[11]={10, 15, 45, 56, 67, 70, 80, 90, 100, 110, 120 }
- 3 Sort the following array using bubble sort: 5,1,4,2,10.

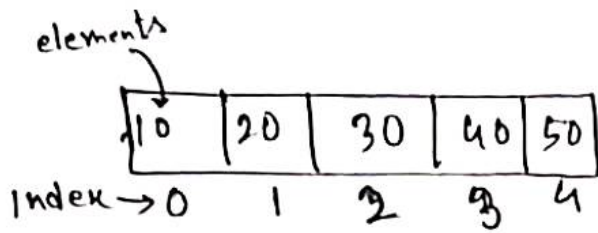
4

3

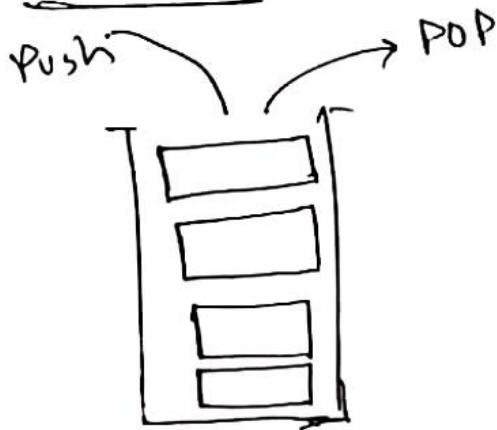
3

① Different types of data structures:

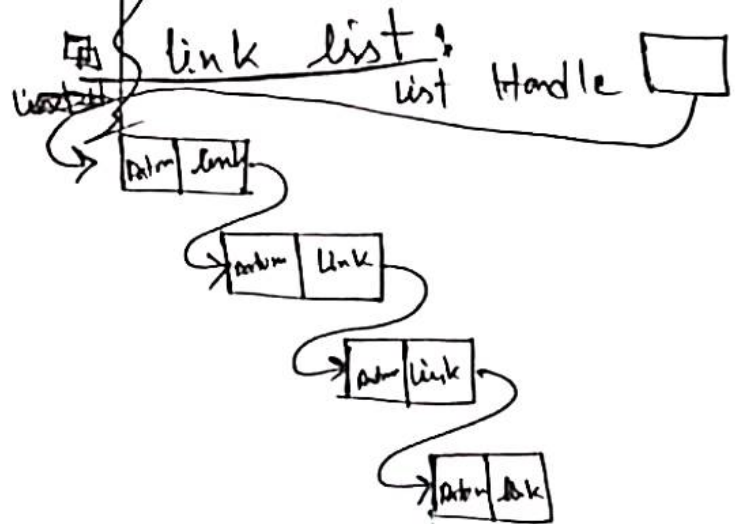
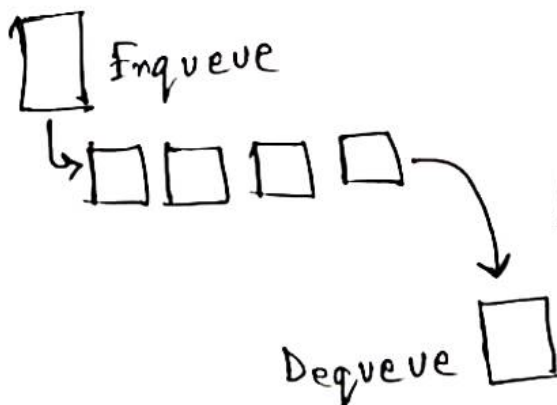
array:



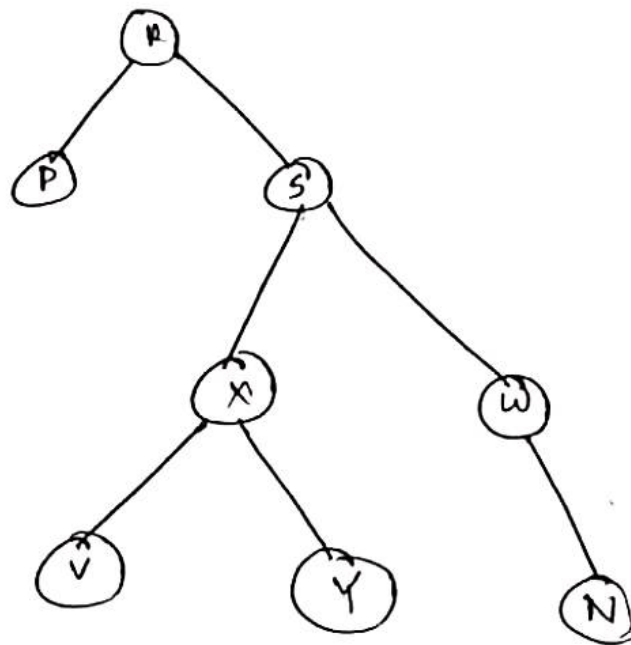
Stack:



Queues:



tree:



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(2)

ITEM = 110

10	15	45	56	67	70	80	90	100	110
0	1	2	3	4	5	6	7	8	9
0								120	
0								10	
0								10	

Int First index = 0

Int last index = 9

Int Middle index = $(0+10)/2 = 5$

Middle value is 70.

if (Middle value == 110) {
 print (Middle value);
}

ITEM = 110

0	1	2	3	4	5	6	7	8	9	10
10	15	45	56	67	70	80	90	100	110	120
					first					last

else {

Now, First index = 5

last index = 10

Int Middle index = $(5+10)/2$

= 7 50, Middle value = 90

if (Middle value == 110) {
 print (Middle value);
}



③ bubble sort:

5, 1, 4, 2, 10

Pass 1:

5, 1, 4, 2, 10

1, 5, 4, 2, 10

1, 4, 5, 2, 10

1, 4, 2, 5, 10

1, 4, 2, 5, 10

Pass 2:

1, 4, 2, 5, 10

1, 4, 2, 5, 10

1, 2, 4, 5, 10

So, the answer is,

1, 2, 4, 5, 10.

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